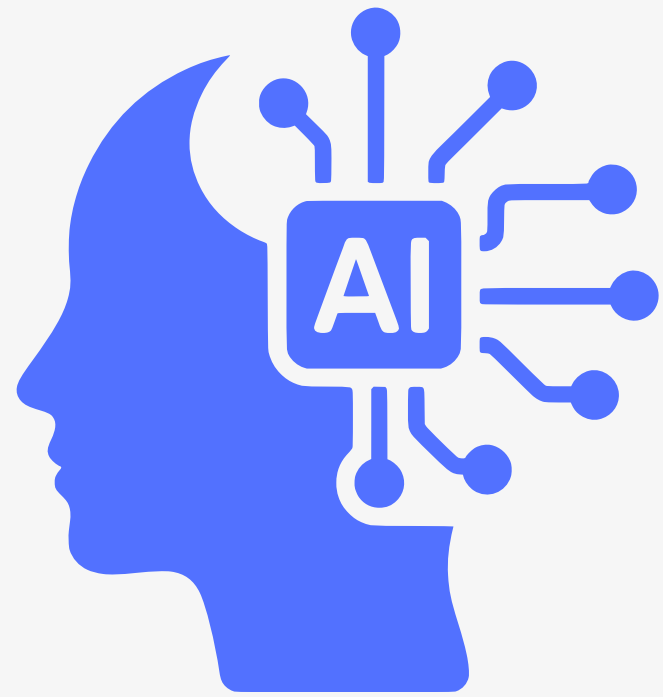




VECTOR DATABASES AT SCALE

AN AI MODEL WITHOUT A VECTOR DATABASE IS LIKE GOOGLE
WITHOUT SEARCH.





WHAT IS A VECTOR DATABASE?

- Stores vector embeddings instead of plain text
- Enables semantic (meaning-based) search
- Finds similar information instantly
- Core component of modern RAG systems





WHY NOT USE SQL DATABASES?

- SQL searches exact keywords
- Vector DB searches by meaning
- Better for AI applications
- Faster semantic retrieval





WHAT ARE EMBEDDINGS?

- Numerical representation of text
- Captures semantic meaning
- Similar meanings have similar vectors
- Used for intelligent search





HOW A VECTOR DATABASE WORKS

- Convert text into embeddings
- Store embeddings
- Convert user query into embedding
- Find nearest matching vectors
- Return relevant chunks





CHROMA DB

- Open-source vector database
- Easy to set up
- Runs locally
- Great for learning and development
- Ideal for small to medium AI projects





PINECONE

- Fully managed cloud vector database
- Highly scalable
- Production-ready
- Handles millions of vectors
- Used by enterprise AI applications





CHROMADB VS PINECONE





METADATA FILTERING

- Filters search results using metadata
- Search only relevant documents
- Improves retrieval accuracy
- Reduces unnecessary AI context





REAL-WORLD APPLICATIONS

- ChatGPT Knowledge Bases
- Enterprise AI Search
- Legal AI
- Medical AI
- Customer Support AI
- Internal Company Documentation

